GradientCIO

PRIVATE MARKETS PORTFOLIO CONSTRUCTION

Northstar Private Markets Program

Portfolio construction review · prepared for the Investment Committee

AS OF
2026-06-30

PRIVATE-MARKETS NAV
$230.0M

UNFUNDED COMMITMENTS
$92.0M

REVIEW
Program construction

01 Executive Decision

Recommendation and committee action

Pacing and concentration need monitoring

RECOMMENDATION

We recommend a two-year $80.0M commitment program that shifts new commitments toward infrastructure and secondary private equity while keeping private-markets exposure inside the 15.0%–30.0% policy range S3 S6. Committee action requested: approve the program ceiling and first-year pacing, subject to manager diligence and quarterly liquidity tests S3 S7.

PRIVATE-MARKETS WEIGHT	UNFUNDED COMMITMENTS	LIQUID COVERAGE	POLICY STATUS
23.0%	$92.0M	4.8x	Near limit
Policy range 15.0%–30.0% S3	9.2% of NAV S4	Governed base case S3	Manager concentration S3

Recommendation and action requested

We recommend a two-year $80.0M commitment program. Committee action requested: approve a $45.0M first-year ceiling and require completed diligence before each commitment S3 S6 S7.

DECISION SUMMARY

DECISION ELEMENT	PROPOSED ACTION	EVIDENCE	STATUS
Program size	$80.0M over two years	S6	Proposed
First-year ceiling	$45.0M	S6	Proposed
Policy fit	Remain within 15.0%–30.0%	S3	Compliant

Conditions and limits

Each manager requires completed operational and GIPS diligence; quarterly pacing proceeds only while governed liquidity coverage remains at or above the returned policy threshold S3 S7.

Key judgment — Pacing

Observation: Commitment pacing and manager concentration both need monitoring S1. Why it matters: The $45.0M first-year ceiling keeps the program inside its liquidity limits. Uncertainty: Capital calls depend on manager deployment speed. What would change the view: Calls running ahead of the pacing plan.

Key judgment — Sleeve size

Observation: Private markets are 23.0% of total NAV S2. Why it matters: New commitments are sized against this weight and the unfunded balance. Uncertainty: Private valuations lag public markets. What would change the view: A valuation update that moves the weight by more than 2pp.

Key judgment — Policy limit

Observation: The private-markets allocation is near its policy limit S3. Why it matters: Each new commitment must be checked against the limit before signing. Uncertainty: The limit test uses lagged valuations. What would change the view: Distributions that bring the weight back toward target.

02 Mandate and Construction Objective

Policy and program constraints

Portfolio	**Northstar Total Fund** S2
Objective	**Diversified private-markets program** S7
Horizon	**10 years** S5
Target range	**15.0%–30.0% of total NAV** S3
Base currency	**USD** S2
Data scope	**Illustrative sample data** S1 S1

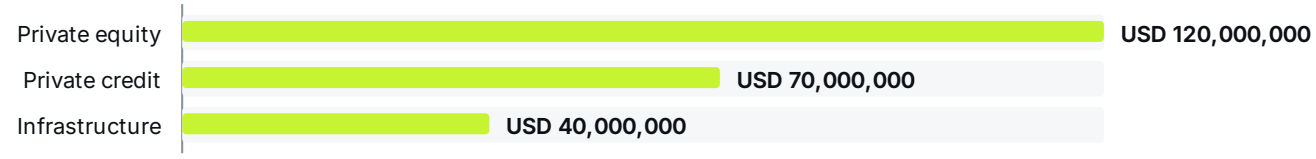
GOVERNED FRAMEWORK

AREA	OBJECTIVE OR LIMIT	OBSERVED	STATUS	SOURCE
Private markets	15.0%–30.0%	23.0%	Compliant	S3
Single manager	At or below 12.0%	11.4%	Near limit	S3 S4
Unfunded ratio	At or below 12.0%	9.2%	Compliant	S3

03 Current Private Markets Portfolio

NAV, commitments and valuation dates

PRIVATE-MARKETS NAV BY STRATEGY



CURRENT PROGRAM SNAPSHOT

STRATEGY	NAV	UNFUNDED	COMMITMENT	% NAV	VALUATION DATE
Private equity	$120.0M	$42.0M	$210.0M	12.0%	2026-03-31 S4
Private credit	$70.0M	$30.0M	$125.0M	7.0%	2026-03-31 S4
Infrastructure	$40.0M	$20.0M	$75.0M	4.0%	2026-03-31 S4

ILPA COMPANY COVERAGE

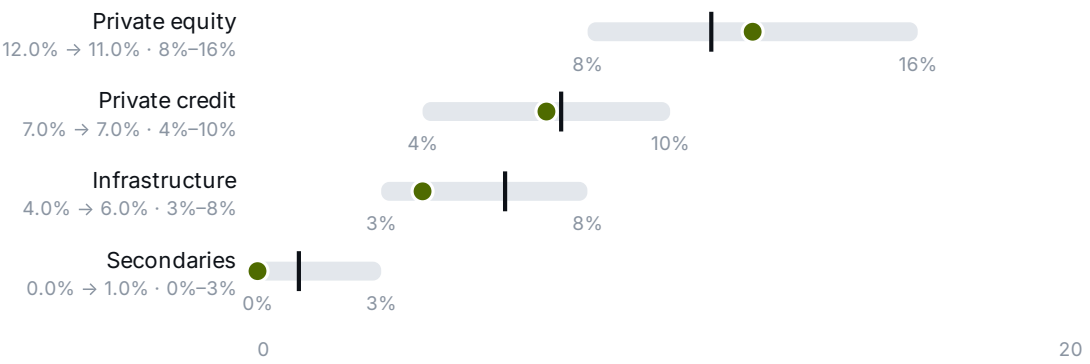
DATA STATUS	MAPPED COMPANIES	VALUED COMPANIES	ASSESSED / TOTAL	AGGREGATE COMPLETENESS	SELECTED ARTIFACTS	REPORTING DATE	DRAFT LABEL
Incomplete	2	2	2 / 3	Incomplete — one unknown-grain holding excluded	1	2026-06-30	Draft — July 22, 2026 S9

All reported private-market NAVs are dated 2026-03-31 and therefore lag the report date; unfunded amounts are shown separately and are not included in NAV. Company look-through is a non-additive risk overlay; the partial server aggregates assess two of three holdings, with one unknown grain excluded S4 S9.

04 Target Portfolio Structure

The proposed mix reduces concentrated risk while preserving mandate exposure

CURRENT WEIGHT VERSUS POLICY RANGE



CURRENT AND TARGET MIX

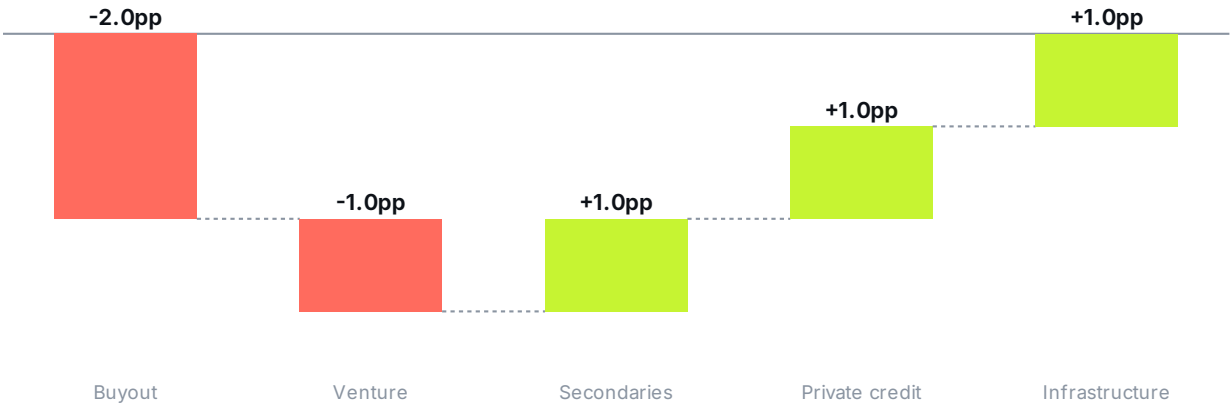


VINTAGE-YEAR MIX



Governed vintage evidence and proposed commitment pacing S6 S7.

TARGET ALLOCATION CHANGES (PERCENTAGE POINTS)



CURRENT VERSUS TARGET

SEGMENT	CURRENT	TARGET	CHANGE	RANGE / CONSTRAINT	RATIONALE
Private equity	12.0%	11.0%	-1.0pp	8.0%–16.0%	Use slower new pacing S3 S6
Private credit	7.0%	7.0%	0.0pp	4.0%–10.0%	Maintain income role S5 S7
Infrastructure	4.0%	6.0%	+2.0pp	3.0%–8.0%	Direct new commitments here S5 S6
Secondaries	0.0%	1.0%	+1.0pp	At or below 3.0%	Add vintage diversification S6 S7

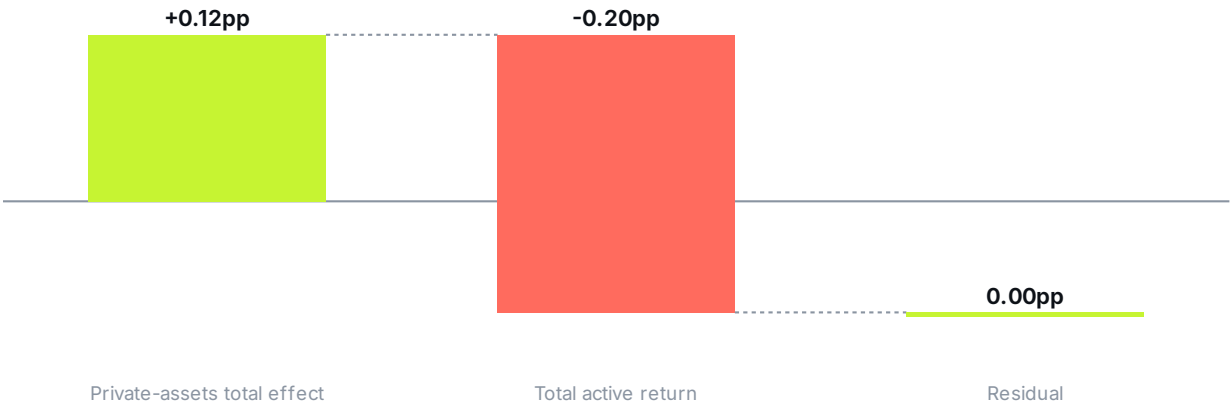
> **Analysis — Why this target**
>
> Observation: New pacing increases two roles, infrastructure and secondaries, versus the current mix S6. Why it matters: The broader mix remains inside policy boundaries S3. Uncertainty: CMA, regime and manager-finding coverage remain incomplete S7. What would change the view: Liquidity coverage below policy or completed records rejecting either role S3 S7.

Targets are a committee proposal from the illustrative mandate and returned policy headroom; they are not changes saved to GradientCIO S3 S7.

05 Historical Performance and Attribution

Returned portfolio evidence; sleeve gaps remain explicit

GOVERNED ATTRIBUTION EFFECTS



Returned attribution effects; residual is shown explicitly S5.

STANDARD PERIODS

PERIOD	PORTFOLIO / SLEEVE	BENCHMARK	EXCESS	COVERAGE
1Y total portfolio	7.2%	7.6%	-0.4pp	Complete S8
3Y total portfolio (ann.)	6.8%	6.5%	+0.3pp	Complete S8

ATTRIBUTION EFFECTS

EFFECT	CONTRIBUTION	SOURCE
Private-assets total effect	+0.12pp	S8
Total active return	-0.20pp	S8
Residual	0.00pp	S8

Governed attribution method and reconciliation metadata follow S5.

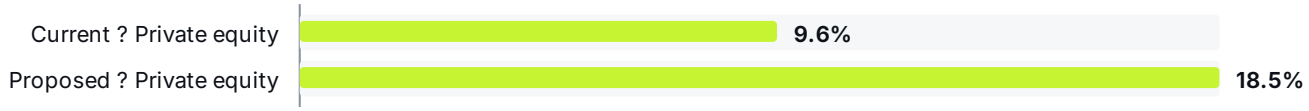
METHOD

Method	**Brinson-Fachler** S8
Linking	**Symmetric Carino** S8
Basis	**Net, monthly linked** S8
Tolerance	**1 bp; reconciled** S8

06 Forward Return and Risk

Governed assumptions frame tradeoffs rather than forecasts

FORWARD EXPECTED-RETURN COMPARISON



Governed forward assumptions; assumptions, not forecasts S6.

Assumption set	**2026 Strategic CMA** S5
Regime	**Baseline** S5
Horizon	**10 years** S5
Currency	**USD** S5
Context fingerprint	**test-pm-forward-2026** S5

PRIVATE-MARKET FORWARD ASSUMPTIONS

SEGMENT	EXPECTED RETURN	EXPECTED VOLATILITY
Private equity	9.6%	18.5%
Private credit	7.1%	9.2%
Infrastructure	7.8%	11.8%

Analysis — Assumptions frame tradeoffs

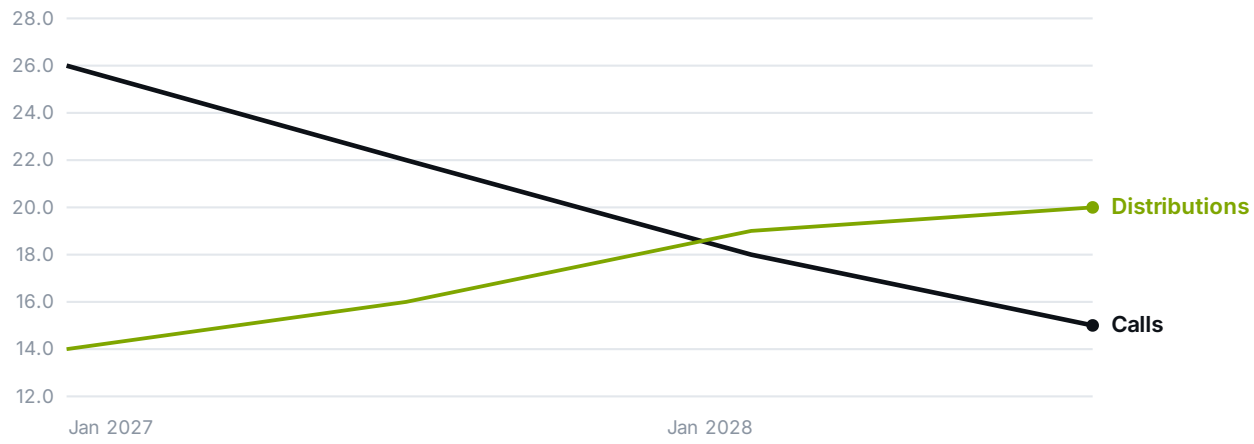
Observation: Private equity has higher expected return and volatility than every other selected segment S5. Why it matters: Segment sizing balances return potential against modeled risk. Uncertainty: These long-horizon assumptions are not manager forecasts. What would change the view: A governed assumptions refresh reversing either relative ranking S5.

The figures are long-horizon assumptions under the selected context, not forecasts or guarantees S5.

07 Commitments, Pacing and Cash Flow

Returned portfolio commitment schedule

COMMITMENT PACING (USD MILLIONS)



Illustrative projection for committee review S6.

PROJECTED CALLS AND DISTRIBUTIONS

PERIOD	CALLS	DISTRIBUTIONS
2026 H2	USD 26,000,000	USD 14,000,000
2027 H1	USD 22,000,000	USD 16,000,000
2027 H2	USD 18,000,000	USD 19,000,000
2028 H1	USD 15,000,000	USD 20,000,000

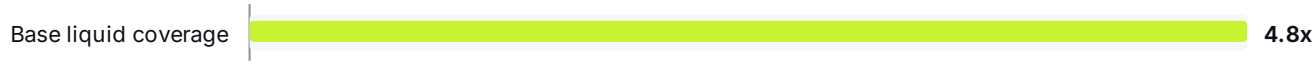
NEAR-TERM PACING

PERIOD	CALLS	DISTRIBUTIONS	NET CASH FLOW	BASIS	SOURCE
2026 H2	$26.0M	$14.0M	-$12.0M	Returned projection	S6
2027 H1	$22.0M	$16.0M	-$6.0M	Returned projection	S6
2027 H2	$18.0M	$19.0M	+$1.0M	Returned projection	S6

08 Liquidity and Denominator Risk

Coverage remains adequate under the sample stress case

GOVERNED SCENARIO COVERAGE



Denominator stress is unavailable S3 S6.

LIQUIDITY COMPOSITION



LIQUIDITY PROFILE

BUCKET	% NAV	AMOUNT	REQUIREMENT	STATUS
Daily to quarterly	44.0%	$440.0M	At least 25.0%	Compliant
Over two years / locked	23.0%	$230.0M	At or below 30.0%	Compliant

Liquid assets	**$440.0M** S3
Unfunded commitments	**$92.0M** S4
Base coverage	**4.8x** S3
Near-term projected calls	**$48.0M** S6

Analysis — Coverage supports pacing

Observation: Base liquid coverage is 4.8x against unfunded commitments S3 S4. Why it matters: Current resources support the staged pacing case. Uncertainty: A governed denominator-stress result remains unavailable for this portfolio. What would change the view: Lower resources, higher calls, or stress results below the policy threshold S3 S6.

SUPPORTED SCENARIOS

SCENARIO	LIQUID RESOURCES	FUNDING NEED	COVERAGE	STATUS
Governed base case	$440.0M	$92.0M	4.8x	Compliant
Denominator stress	Not available — no governed stress result	Not available	Not available	Not assessed

09 Diversification and Manager Diligence

Concentration and pre-commitment conditions

DIVERSIFICATION

AXIS	LARGEST EXPOSURE	SHARE	LIMIT	STATUS	SERVER METRIC / BASIS
Manager	Example Private Manager A	11.4%	12.0%	Near limit	Commitment exposure; additive S4
Strategy	Private equity	52.2% of private NAV	Not assessed	Not assessed	Commitment exposure; additive S4
Portfolio company	illustrative Company A	24.0% of valued company NAV	Not assessed	Partly available	Server company_concentration; canonical NAV; non-additive S9
Sector	Technology	41.0% of valued company NAV	Not assessed	Partly available	Server sector_concentration; canonical NAV; non-additive S9

PORTFOLIO-COMPANY EVIDENCE

COMPANY	CANONICAL NAV	EXACT ILPA VALUE	VALUATION / REPORTING DATE	PROJECTED RANGE	LEVERAGE / COVERAGE	COVENANT STATUS	OPERATING-QUALITY EVIDENCE	RECONCILIATION	SOURCE
illustrative Company A	$24.0M	FMV $24.0M in USD	2026-06-30 / 2026-06-30	Leverage 3.0–4.0x	Interest coverage 2.0–3.0x	In compliance	Revenue growth 5%–10%	Matched	S9
illustrative Company B	$18.0M	FMV $18.0M in USD	2026-06-30 / 2026-06-30	Leverage 4.0–5.0x	Interest coverage 1.5–2.0x	On watch list	EBITDA margin 15%–20%	Matched	S9

Selected artifact illustrative-ilpa-artifact-test, version 1; parser ilpa_portfolio_company_draft_2026_07_22 version 1.0.0; projection registry private-markets-holding-exact-to-range-v1, checksum aaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaa. Draft — July 22, 2026. Canonical values, exact ILPA source values and projected Gradient ranges remain separate S9.

MANAGER EVIDENCE

MANAGER / FUND	EXPOSURE	DILIGENCE STATUS	OPEN FINDINGS	NEXT STEP
Example Private Manager A	$114.0M	Current S7	1 Medium S7	Close valuation-control follow-up
Proposed infrastructure manager	Not funded	Not assessed S7	Not assessed	Run manager comparison and ODD

10 Proposed Program and Implementation

Phased commitments with decision gates

COMMITMENT PROGRAM

PERIOD	SEGMENT	PROPOSED AMOUNT	FUNDING SOURCE	CONDITION	SOURCE
2026 H2	Infrastructure	$25.0M	Distributions and liquid reserve	Complete ODD	S6 S7
2027 H1	Private credit	$20.0M	Distributions	Coverage remains compliant	S3 S6
2027 H2	Secondaries	$20.0M	Distributions and cash	Committee checkpoint	S3 S6
2028 H1	Reserve	$15.0M	To be determined	Evidence refresh	S6

ALTERNATIVES CONSIDERED

ALTERNATIVE	BENEFIT	COST / RISK	REASON NOT SELECTED
No new commitments	Maximum liquidity	Target exposure declines	Does not address program objective S3 S6
Front-load commitments	Faster target progress	Higher near-term calls	Coverage sensitivity S3 S6

- Phase each commitment after diligence clearance S7.
- Refresh pacing and policy liquidity quarterly S3 S6.
- Return to committee if manager concentration reaches its governed limit S3.

11 Risks, Open Items and Approvals

Conditions that govern implementation

DECISION RISKS

PRIORITY	RISK	EVIDENCE	MITIGATION / CONDITION	OWNER
High	Valuation and denominator lag	Q1 valuations S4	Quarterly headroom test	Investment staff
Medium	Manager concentration	11.4% vs 12.0% S3 S4	Cap follow-on commitments	CIO
Medium	Near-term net calls	$18.0M through 2027 H1 S6	Maintain liquidity gate	Treasury

OPEN ITEMS AND APPROVALS

TIMING	ITEM	REQUIRED EVIDENCE	OWNER	STATUS
Before approval	Confirm first manager	Comparison and ODD	Private markets team	Open S7
Within 30 days	Document pacing limits	Final program schedule	CIO	Open S6
Next review	Refresh valuations	2026 Q2 NAVs	Operations	Open S4

12 Coverage

Portfolio and allocation	**Available**	Illustrative sample data S2
Policy	**Available**	Allocation, liquidity and concentration S3
Exposure	**Available**	NAV, commitments and unfunded S4
ILPA portfolio-company look-through	**Incomplete**	2 of 3 holdings assessed; one unknown grain; Draft — July 22, 2026 S9
Forward assumptions	**Available**	2026 Strategic CMA S5
Commitments charts	**Available**	Returned pacing projection S6
Manager diligence	**Incomplete**	Proposed manager not assessed S7
Returns and attribution	**Incomplete**	No sleeve-specific return S8
CMA consensus	**Not available**	Optional context not returned S5
Macro regime	**Not available**	Optional context not returned S5
Strategy Lab	**Not available**	No selected series S8
Peer allocation context	**Not available**	Optional context not returned S7
Transaction costs	**Not available**	Not returned S8

A Appendix — Sources and methods

Based on 9 of 10 evidence areas.

TAG	EVIDENCE	SOURCE	AS OF	CHECK
S1	Org capabilities	list_organizations; get_gradient_capabilities	2026-06-30	Passed
S2	Portfolio / allocation	list_portfolios; get_portfolio_structure	2026-06-30	Passed
S3	Policy assessment	check_portfolio_policy	2026-06-30	Passed
S4	Private-market exposure	get_portfolio_exposure	2026-03-31	Passed
S5	Forward assumptions	get_capital_market_assumptions	2026-06-15	Passed
S6	Commitments and pacing	get_portfolio_exposure	2026-06-30	Passed
S7	Manager / mandate	get_gradient_capabilities	2026-06-30	Advisory
S8	Returns / attribution	get_portfolio_historical_returns; get_portfolio_attribution	2026-06-30	Incomplete
S9	ILPA company look-through	get_private_markets_holdings	2026-06-30	Partly available

Illustrative, Gradient Maintained — demo data, not the client's holdings or managers. All organizations, portfolios, managers and results in this public example are illustrative.

SOURCE AUDIT TRAIL

TAG	KEY PARAMETERS	DIGEST
S1	org=test	sha256:aaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaa
S2	allocation_tree	sha256:aaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaa
S3	portfolio=test	sha256:aaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaa
S4	all pages	sha256:aaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaa
S5	baseline	sha256:aaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaa
S6	commitments	sha256:aaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaa
S7	illustrative	sha256:aaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaa
S8	all sections	sha256:aaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaa
S9	paged; server aggregates	sha256:aaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaa

Data scope for every source: Illustrative sample data. Each digest is the SHA-256 fingerprint of the exact data the report used.

B Appendix — Server Metric Methods and Disclosures

SERVER METRIC METHODS

SOURCE	QUANTITY	METHOD / FORMULA	VERSION	BASIS / UNITS / TOLERANCE
S3	Policy status	Governed portfolio assessment	portfolio-policy-v1	Returned status
S4	Exposure	Server exposure aggregation	portfolio-exposure-v1	USD; unfunded separate
S5	Forward statistics	Expected-statistics charts	chart-contract-v1	10-year assumptions
S9	Company concentration	Server aggregates only	private-markets-holding-exact-to-range-v1	Canonical NAV; non-additive; partial coverage

Private-market valuations may lag the report date S4. Company look-through is non-additive to commitment exposure, and exact ILPA source money is not summed across currencies S9. The selected portfolio-company artifact is Draft — July 22, 2026, not a final ILPA release S9. Forward values are assumptions, not forecasts or guarantees S5. Portfolio Analytics and Strategy Lab are separate analytical scopes. This AI-assisted construction analysis is not investment, legal or compliance advice.